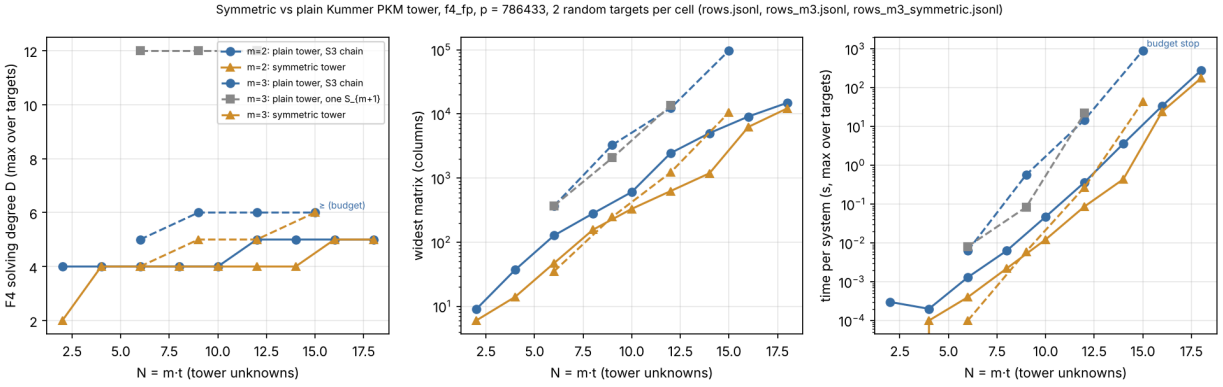
Symmetric PKM tower

The symmetric PKM tower: F4 solving degree against the plain Kummer tower

2026-10-26 · item 1 of the 2026-10-26 triage (`docs/session-logs/2026-10-26.md`) · instrument `examples/pkm_symmetric_tower.rs` · records `rows.jsonl` (m = 2), `rows_m3.jsonl`, `rows_m3_symmetric.jsonl` (m = 3), `run.log` · graph `d_vs_n.svg` (`plot.py`, recorded values only) · PDF `report.pdf`

Class: stage diagnostic of the solver axis (AGENTS.md §2). No pipeline is priced, no S, no leaderboard row. It answers one question about the factor-base encyclopedia's flagship tower family (`docs/ic/factor-base-encyclopedia/README.md` §0.1, §8): does combining the Kummer membership and the point-sum constraint in the elementary symmetric functions, *before* expansion, change the F4 solving degree?



Verdict

Yes, by one degree, and with matrices about 10× narrower at m = 3. The growth of `D` with `N` is not removed. The symmetric tower's `D` still rises (4, 5, 5, 6 at `m = 3`, `N` = 6, 9, 12, 15). The effect is a shift of the staircase, as far as these cells can show, not a different law. It does not settle lead 1, whether `D` stays bounded as `N` grows.

m	N	plain chain D	plain, one S_4 D	symmetric D	widest matrix, plain / sym	time per system, plain / sym
2	10	4	—	4	606 / 331	0.05 s / 0.01 s
2	12	5	—	**4**	2,451 / 627	0.36 s / 0.09 s
2	14	5	—	**4**	5,031 / 1,188	3.6 s / 0.44 s
2	16	5	—	5	9,119 / 6,281	34 s / 24 s
2	18	5	—	5	14,922 / 12,214	280 s / 176 s (one target)
3	6	5	12	**4**	368 / 35	< 0.01 s
3	9	6	12	**5**	3,285 / 248	0.6 s / 0.02 s
3	12	6	12	**5**	12,314 / 1,224	14.8 s / 0.3 s
3	15	≥ 6 (900 s budget stop, 97,456 columns)	aborted	**6**	≥ 97,456 / 10,532	> 900 s / 43 s

Values are the maximum over the two random targets of each cell (identical in every cell where both finished). "Plain, one S_4" replaces the S_3 chain by the single summation polynomial S_4 (by resultant). Its D = 12 is S_4's own degree, so a single high-degree `S_{m+1}` without symmetrisation is the worst arm. The recorded tower-engine D for the plain Kummer tower at `m = 3, N = 15` is 7 (encyclopedia §0.1). The plain `f4_fp` run here did not finish within its budget.

Construction

Kummer base `V = g·μ_{2^t}`, summands `x_i = g·w_i`. Distinct `w_i ∈ μ_{2^t}` ⟺ `Q(X) = ∏(X − w_i)` divides `X^{2^t} − 1` ⟺ `X^{2^t} ≡ 1 (mod Q)`. The unknowns are the `e_k` (coefficients of `Q`) and the coefficients of `R_j ≡ X^{2^j} mod Q` for `j = 1..t−1`, so `N = m·t` as in the plain system. The equations are `R_j = R_{j−1}² mod Q`, `R_{t−1}² ≡ 1`, and `S_{m+1}(g w, x_R)` rewritten in the `e_k` (leading-term algorithm). The symmetric system sees unordered summand sets of distinct points only. That is the same relation for index calculus, with `m!` fewer solutions; doublings `x_i = x_j` are not represented.

Controls and checks:

- **Positive control on the instrument.** The plain arm is the pilot's system (`examples/pkm_tower_pilot.rs`, `tower` control). On cells run by both, the pilot binary gives the same D and the same matrix widths (9, 37, 127, 280 columns at N = 2..8).
- **Planted check.** Every system of every arm is first evaluated on a planted decomposition, distinct summands from `V` and an `x_R` solving `S_{m+1}`. Any non-vanishing equation stops the run.
- **Reproducibility.** The symmetric-only rerun (`rows_m3_symmetric.jsonl`) reproduces every symmetric row of `rows_m3.jsonl` exactly (same D, widths, inconsistency).
- Same prime (`p = 786433 = 3·2^18 + 1`), curve, `g`, `ζ` and targets for all arms. The seeds are fixed in the source. Engine `f4_fp`, grevlex, degree bound `n + d + 6`, measured on random targets (`--random 2`), as the recorded Ds were.

What did not run, and why

- `m = 2, N = 18`, target 1, plain chain: ran about 55 minutes and 4.4 GB against a 900 s budget. The operator stopped it, because `f4_fp` checks its budget between steps, not inside one. Its symmetric row was not run (`run.log`).
- `m = 3, N = 15`: the plain chain hit its 900 s budget (D ≥ 6, 97,456 columns). The plain-S_4 arm then aborted (exit 134, an allocation failure under the 10 GB cap) before the symmetric arm ran. The symmetric arm was re-run on its own: the draws do not depend on which arms run.
- `m = 4` and `N > 18` were not attempted on this 4-core, 15 GB container.
- The resume checkpoint could not be pushed: the container has no S3 token.

Reading and next step

The symmetric presentation is a real constant-factor improvement for the decomposition oracle's F4 stage: one degree less and about 10× narrower matrices at `m = 3` for `N ≤ 15`, and about 50× faster per system at `N = 12` (14.8 s against 0.3 s) and more than 20× at `N = 15` (over 900 s against 43 s). It is not an index-calculus improvement until it is priced end to end. Its D keeps growing, so it does not answer lead 1 (bounded D). The next measurement that would matter is the symmetric arm at `m = 3, N = 18, 21`, which is cheap (43 s at N = 15), and at `m = 4`. If D still steps up every one to two cells, the family's subexponentiality question stands exactly as before, one degree lower.